



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll No.....

Name of the Course: BCA/BCA AI & DS

Semester: 2nd Semester

Name of the Paper: Probability and Statistics for Data Science

Paper Code: TBC-204/TBD-203

Time: 1.5 Hours

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1.

(10 Marks)

a. What is Exploratory Data Analysis (EDA)? Explain its types with examples.

(CO1)

OR

b. What are the different types of data, and why are they important?

(CO1)

Q2.

(10 Marks)

a. Describe the methods of collecting data.

(CO1)

OR

b. What is meant by 'understanding the business problem' in data science?

(CO1)

Q3.

(10 Marks)

a. State Baye's theorem of probability. In a bolt factory, machines A, B, and C manufacture, respectively 25%, 35% and 40% of the total. Of their output, 5, 4, and 2 percent are defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it was manufactured by machine B?

(CO2)

OR

b. The mean yield per plot of a crop is 17kg, and the standard deviation is 3kg. If the distribution of yield per plot is normal, find the percentage of plots giving yield:

(CO2)

(i) Between 15.5Kg and 20Kg; and

(ii) More than 20kg.

Given that the area under the standard normal curve $Z = 0$ to $Z = 0.5$ is 0.1915, and, $Z = 0$ to $Z = 1$ is 0.3413.

Q4.

(10 Marks)

a. Explain the Poisson distribution. The mean and variance of a Binomial distribution are 4 and 4/3, respectively. Find the probability of,

(CO2)

(i) Exactly two successes,

(ii) More than two successes.

OR

b. Define the multiplication law of probability. An urn contains nine balls, two of which are red, three of which are blue, and four of which are black. Three balls are drawn at random. What is the probability that,

(CO2)

(i) The three balls of different colours,

(ii) The three balls are of the same colour.



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Q5.

(10 Marks)

- a. Define with examples.
(i) Mutually exclusive cases,
(ii) Independent cases,
(iii) Equally likely cases,
(iv) Random experiment,
(v) Sample space.

(CO2)

OR

- b. Define the addition law of probability. An urn contains 10 black balls and 10 white Balls. Find the probability of drawing two balls of the same colour.

(CO2)